

Timothy Kim

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EDUCATION

Emory University

Atlanta, GA

Applied Mathematics and Statistics, Double Major: Computer Science

Cumulative GPA: 3.9/4.0

EXPERIENCE

Emory University AI.Data Lab

Atlanta, GA

Student Research, Sports + Data Analytics

Sep. 2025 - Present

- Visualized 2,500,000+ NFL positional data points spanning 5+ years with Matplotlib for CNN model input
- Developed a convolutional neural network, generated Grad-CAM heat map, predicting high-impact running plays and visualizing most predictive on-field areas for yardage gain
- Reached ~80% accuracy in predicting yardage outcome within 5 yards

South Louisiana Swim Team

New Orleans, LA

Swim Instructor

Jun. 2024 - Aug. 2025

- Taught swim lessons and led pre-competition training practices for children (ages 2–14), covering water safety, life-saving basics, and advanced stroke development. Prepared athletes for higher-level groups and competitions
- Maintained pool facilities, ensuring safe chemical levels, organized gear, and a clean training environment.

PROJECTS

Swim Performance Forecasting | *BeautifulSoup/Selenium, scikit-learn, Streamlit*

- Built an end-to-end Python pipeline to scrape and process 100,000+ Swimcloud race records, anonymize swimmer identifiers, and extract each swimmer's annual best times.
- Achieved a 23% reduction in prediction error using a Gradient Boosting model benchmarked against baseline predictors to forecast junior-year times from previous years
- Developed an interactive Streamlit dashboard to forecast best-, mid-, and worst-case future swim times, converting model outputs into user-facing athlete development scenarios.

Trouble Strategy Simulator | *feature-based reinforcement learning, Monte Carlo simulation, Python*

- Developed a Python simulation engine for the board game Trouble, supporting strategy assignment and bot gameplay, and batch Monte Carlo experiments
- Implemented heuristic and reinforcement-learning agents, including a softmax policy-gradient bot trained on move features such as captures, danger, and finishing pieces.
- Trained RL bot to win ~30% of four-player games against near-optimal heuristic agents, outperforming the 25% baseline expectation

NBA Lead Tracker Visualization | *React, Vite, FastAPI, Python, Pandas*

- Built full-stack visualization app that animates NBA game score progression using nba_api data
- Developed a FastAPI backend to fetch, clean, and process NBA play-by-play data into point-differential timelines using Python and Pandas
- Designed interactive React/Vite frontend with custom playback controls, speed adjustment, timeline navigation, and responsive UI.

SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS

Data/ML: pandas, NumPy, scikit-learn, Matplotlib, Gradient Boosting, CNNs, model evaluation, feature engineering

Web/Deployment: React, FastAPI, Streamlit

Tools: Git, GitHub, Jupyter Notebook, Excel